

Muhammad Fawad Shah

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EDUCATION

University of Engineering and Technology (UET) Peshawar
Bachelor of Science in Computer Systems Engineering

2024 - 2028

OPEN SOURCE CONTRIBUTIONS

Haiku OS — exFAT Filesystem Write Support

C++ · Kernel · VFS

- Engineered native write support framework within Haiku OS kernel, implementing Virtual File System abstraction layer, block cache integration, and inode management for full exFAT read-write capability.
- Architected FAT allocation management with dynamic bitmap tracking and validation routines ensuring filesystem integrity under high-concurrency I/O operations.
- Developed bidirectional UTF-16 to UTF-8 character encoding conversion with comprehensive error handling for POSIX-compliant directory traversal.
- Mainlined into production codebase after peer review by core maintainers; 20+ files, 333+ additions. [\[View\]](#)

PROJECTS

High Performance Limit Order Book

C++ · Low-Latency · [\[GitHub\]](#)

- Designed ultra-low-latency financial matching engine using custom data structures achieving $O(1)$ order insertion, deletion, and lookup operations.
- Optimized execution pipeline to allow sub-microsecond latencies via compile-time execution and contiguous allocation strategies.

StarPen — Full-Stack Blogging Platform

Flask · PostgreSQL · SQLAlchemy · [\[Live\]](#)

- Architected production-grade blogging system with secure session authentication, token-verified password reset, and CSRF protection across all endpoints.
- Designed normalized PostgreSQL schema with optimized indices and nested relational queries supporting efficient multi-tier comment tree retrieval.
- Used PostgreSQL in development and Supabase in production.

LeetCrack — AI-Powered Programming Assistant

Flask · LLM APIs · [\[GitHub\]](#)

- Built automated data ingestion pipeline with custom API client to index 3,000+ competitive programming problems into persistent local storage.
- Integrated LLM orchestration via Gemini and Llama APIs with structured prompting and function-calling for algorithmic code generation and complexity analysis.

File Compression Engine — Huffman Encoding

C++ · Binary I/O · [\[GitHub\]](#)

- Implemented high-performance compression utility using Huffman Coding achieving ~50% compression on text inputs.
- Engineered efficient binary serialization/deserialization with full lossless recovery validation.

TECHNICAL SKILLS

Languages: C++, Python, C, JavaScript, SQL, Bash, HTML/CSS

Systems & Kernel: POSIX APIs, Linux/Unix internals, Kernel development, Virtual Memory, File Systems, Process Management

Web: Flask, SQLAlchemy ORM, Jinja2, Bootstrap, Tailwind CSS, RESTful APIs

Databases: PostgreSQL, Supabase, SQLite, Query optimization, Database design

Tools: Git/GitHub, GCC/Clang, GDB/Valgrind, CMake, Shell scripting, GitHub Actions

Specializations: Data Structures & Algorithms, Low-Latency Architecture, Cache Optimization, API Integration

COMPETITIVE PROGRAMMING

- Active competitive programmer solving algorithmic problems on LeetCode and Codeforces; specializing in graph theory and dynamic programming.